TaskFlow Tech

HABITBUDDY

TEST PLAN

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Introduction

The Test Plan has been created to communicate the test approach to team members. It includes the objectives, scope, schedule, risks and approach. This document will clearly identify what the test deliverables will be and what is deemed in and out of scope.

1.1 Objectives

HabitBuddy is a web-based mobile application designed to help children aged 6-12 build and maintain positive habits. The system empowers the parents to create, customize and monitor habits while the children are motivated through a gamified experience involving virtual coins that can be redeemed for rewards, and badges that can be earned for completing certain milestones. The primary objective of this system is to create an engaging tool that will give parents a supportive framework for habit formation in their children.

Sprint 1 of the project will deliver the core layout and structure of our application. The key wireframes that will be included are parent registration/login, main child view, habit and step creation, habit and step completion, reward store, badge gallery, add/remove rewards, add/remove badges. This does not represent all the screens that will exist, but it provides a good idea of the core functionality of our application.

1.2 Team Members

Resource Name	**Role *(examples are given below)***
Rishav Dahal, Julian Bertrand, Abyan Hu, Derian Mendez	Developer/Tester
Rowland Foster	Project Manager / Tester/UX Design, Developer

2 Scope

The initial sprint will include ‘must have’ requirements. These and any other requirements that get included must all be tested.

The following sections indicate what is tested during each sprint. The scope of testing is determined at the beginning of the current sprint.

At the end of Sprint 1, a user must be able to:

1. Create a new habit (R1)
2. Define Habit Steps (R2)
3. Mark Step as Complete (R4)
4. View Parent Dashboard (R10)

Initial testing will be conducted to ensure each screen functions as intended. User testing from third-parties will be utilized to gather feedback on current layout and features.

Assumptions / Risks

2.1 Assumptions

1. All team members will have access to the necessary development and testing environments
2. The project requirements will remain the same throughout the given sprint.
3. The application will first be tested in a web browser and later be ported to a mobile web application.
4. The technologies selected for the project will function as expected.

2.2 Risks

The following risks have been identified and the appropriate action identified to mitigate their impact on the project. The impact (or severity) of the risk is based on how the project would be affected if the risk was triggered. The trigger is what milestone or event would cause the risk to become an issue to be dealt with.

#	Risk	Impact	Trigger	Mitigation Plan
1	Scope Creep – as testers become more familiar with the tool, they will want more functionality	High	Delays in implementation date	Each iteration, functionality will be closely monitored. Priorities will be set and discussed by stakeholders. Since the driver is functionality and not time, it may be necessary to push the date out.
2	Key feature is more complex than estimated	Medium	Product did not get delivered on schedule	If a certain feature proves to be too difficult or unrealistic, its scope will be simplified or delayed to a later sprint.
3	Lack of diverse physical devices for testing the application usability and compatibility	Low-Medium	UI bugs or non-functionality on some devices	The application will be tested on as many devices as possible to ensure it functions on most mobile devices.
4				
5				

3 Test Approach

The project is using an agile approach, with 3-week sprints. Developers will conduct basic unit tests on their code before committing it in the repository. Once a feature is complete, it will be deployed to our shared testing environment. If any defects are found, they will be noted and assigned back to the developers for resolution.

<u>Sprint 1</u>

The UI functionality will be rigorously tested to ensure it functions as intended. Limited user testing will also be conducted to gather feedback on the layout and any features that should be added.

Sprint 2

Sprint 3

3.1 Test Automation

Automated testing is designed to improve the efficiency and effectiveness of software testing. Popular tools for this purpose include: Selenium, Python PyTest, Playwright, and Cypress. Due to the limited nature of this project, our primary testing approach will be manual. Manual testing allows for exploratory testing which is important for finding usability issues that automated scripts could miss.

3.2 Test Cases (Black Box)

3.2.1 Feature F9

Test Case ID	Description	Requirements Trace	Directions	Expected Output
TC-001	Verify parent can register a new account			
TC-002	Verify parent can log in with valid credentials			
TC-003	Verify login fails with invalid credentials			
TC-004	Verify parent can create a child profile			

3.2.2 Feature F5

Test Case ID	Description	Requirements Trace	Directions	Expected Output
TC-101	Verify parent can create a reward			
TC-102	Verify child earns coins for completing steps			
TC-103	Verify child can redeem a reward from the store			
TC-104	Verify the correct number of coins are deducted from the child's balance			
TC-105	Verify the new coin balance and reward redemption is reflected on the parent dashboard			

3.2.3 Feature F1

Test Case ID	Description	Requirements Trace	Directions	Expected Output
TC-201	Verify parent can create a new habit with steps			
TC-202	Verify child can view assigned habits			
TC-203	Verify child can mark a habit step as complete			
TC-304	Verify parent dashboard reflects progress			

3.3

Test Cases (White Box)

3.3.1..n Name by Feature (like above)

Organize the test cases using the table below. Be sure to have a directory in your repository for your test suite that follows a naming scheme that matches the items in the table below.

Test Case ID	Description	Directions/Goals	Expected Output
TC-001	Input validation for forms	Test various forms (registration, habit creation, badge creation) with empty, invalid, or oversized data.	Forms display an error message if invalid data is entered.
TC-002	No internet connectivity	Test the application behavior when the application has no internet connectivity. Verify the app syncs when connectivity is reestablished.	The application still functions and the new data is still reflected after being synced when internet connectivity is reestablished.
TC-003	Test basic SQL injection	Use SQL injection strings in login and other input fields to ensure they are sanitized	All inputs are validated/sanitized to prevent inputs being misinterpreted as an SQL query.
TC-004			
TC-005			
TC-006			
TC-007			
TC-008			
TC-009			

4 Test Environment

Development Environment: Each team member will work on their local machine. This environment will be used for coding and initial unit testing.

Testing/Staging Environment A single shared environment will be hosted on GitHub, allowing each team member to clone the project to their machine for functional, integration, and regression testing.

5 Test Schedule

Task Name *(sample is below, focus on sprint 1 to start)*	**Start**	**Finish**	**Effort**	**Comments**
Test Planning	10/02/2025	10/14/2025		
Review Requirements documents	10/02/2025	10/14/2025		
Create initial test estimates	10/06/02025	10/14/2025		
Learn new test resources	10/08/2025	10/14/2025		
First deploy to QA test environment				
Functional testing – Sprint 1				
Iteration 2 deploy to QA test environment				
Functional testing – Sprint 2				
System testing				
Regression testing				
Usability Testing				
Resolution of final defects and final build testing				
Deploy to Staging environment				
Performance testing				
Release to Production				